

A MODULAR ARCHITECTURE SUPPORTING MULTIPLE HYPOTHESES FOR CONVERSION OF TEXT TO PHONETIC AND LINGUISTIC ENTITIES

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ABSTRACT

In this communication we devise a distributed modular scheme for organizing the different types of knowledge needed in the first phase of text-to-speech conversion, namely the conversion of the input text to a symbolic notation, representing phonetic transcription together with syntactic, semantic and pragmatic information. We argue that the proposed scheme will provide for easier formulation and maintenance of knowledge, and that its modularity will also simplify the implementation of different modalities, e.g. various reading modes.

Keywords: Text-to-Speech Conversion, TTS, speech synthesis, knowledge representation

BACKGROUND

Developing and maintaining large knowledge bases to which several people are allowed to contribute over a long period of time is a problem with which we are confronted in our daily work with Text-to-Speech (TTS) systems for more than 10 different languages. The first "half" of TTS conversion, which could be described as "text-to-phonetics", involves knowledge from several domains, each of which is a research area of its own, e.g. lexicography, morphology, syntactic, semantic and possibly pragmatic analysis, but also less formalized knowledge regarding typing conventions etc. An excellent tutorial, indicating the various problems associated with text analysis and word pronunciation appears in Liberman and Church [9]. Formalizing this knowledge is part of the problem. The other part of the problem is that, traditionally, most TTS systems have had a very rigid, sequential structure, where the different modules can not easily share each other's results and where each module's textual scope (or "window") is predefined, and too narrow in some cases.

Typically, in order to arrive at a transcription of a sentence, the transcription at the word level is first determined using a user lexicon, a pre-processor for the handling of abbreviations and digits, an exceptions lexicon and/or grapheme-to-phoneme conversion rules. Finally, sentence level phonological rules are applied. There is often no support for propagation of several hypotheses through the modules, which leads to errors due to decisions being made too early.

SYSTEM ARCHITECTURE

While the problem of properly converting text to phonetic transcription is traditionally dealt with by fitting a context-sensitive text normalizer or pre-processor as a front end, we argue that this problem is better solved *within* the system, by dividing existing knowledge into modules or knowledge sources (KS:s), and letting these KS:s share each other's results, something which provides for incremental problem-solving. We therefore propose using an architecture with distributed modules, which are allowed to interact only via a well-defined data structure.

In this paper some of the knowledge sources, in various stages of development, are described together with the data structure. We will point to the interdependencies between modules as a motivation for the proposed non-sequential structure. The control structure that is used to supervise the execution of modules will be the topic of forthcoming communication, and is not treated in this paper. The KS:s are allowed to vary in terms of computational method (rule-based, probabilistic etc.). In this way, knowledge can be expressed using a suitable formalism for each sub-problem, which provides for easier development and maintenance of the knowledge bases, as similarly arranged speech recognition systems showed in the 70's [6]. Another feature of this system structure is that different reading modalities as well as adaption to new text domains can be achieved just by exchanging the knowledge contained in one or more modules, or by turning them off entirely. Furthermore, the proposed

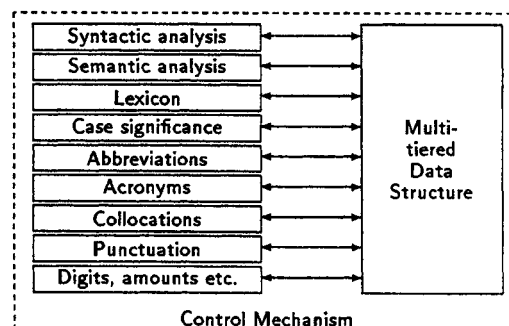


Figure 1: System architecture

scheme allows for propagation of multiple hypotheses, which means that decisions do not have to be made until there is sufficient information to do so.

The proposed scheme is illustrated in Fig. 1. The data structure is a multi-tiered structure consisting of phrases, words, orthography, transcription, syllables, segments (phonemes), and information like parameters and binary features on separate levels.

KNOWLEDGE SOURCES

Typically, knowledge sources cover very small domains. On the other hand they usually perform quite well within that domain. For example, one KS may be good at converting phone numbers into a phonetic transcription, while another is good at zip codes. Some KS:s are capable of supplying multiple hypotheses. Below, some of the KS:s are described.

Punctuation ambiguity

A TTS system can be used in *interactive* mode, e.g. when used in a communication device, where the user *types* the text, and the system converts it to synthetic speech as it is input. The other option is a kind of non-interactive “batch” mode, where you feed the TTS system a chunk of text, e.g. a page of newspaper text. The obvious difference is that, in the latter case, the TTS system can look ahead in the text, whereas in the first case, it can only have access to previously input text.

In non-interactive mode, it seems reasonable that a TTS system should first try to divide the text into sentences. This is not as easy a task as it might seem at first, as shown by e.g. Liberman and Church [9]. Punctuation marks have other usages than ending sentences, e.g. in abbreviations. Capitalization is not a particularly reliable indicator of sentence beginnings either, since it is used in most languages for names, in some for titles, and e.g. in German for nouns. We have investigated the approach suggested by Liberman and Church, which is to tune an algorithm based on pattern-matching to data, in this case excerpts from the Swedish texts of the SUC (Stockholm-Umeå Corpus) [4] covering various fields including newspaper texts, technical reports and novels. These texts display a variety of sentence ending patterns. The error rate of a matching algorithm based on a pattern similar to the one in Fig. 2 is typically below 1 % [5].

There are two other KS:s with which this module is interrelated: 1) the lexicon, in particular the abbreviations and proper noun sub-components and 2) the module dealing with typographic case significance. Including known titles and proper names in the pattern would make it possible to eliminate a number of erroneously detected sentences endings in cases like “Mr. Speech Technology attends the Eurospeech Conference every second year”. For the interactive mode, with its inherent restriction on availability of data, a more conventional approach without look-ahead must be used.

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{not whitespace}{punctuation mark}+{close parenthesis}?
{quote character}?{whitespace or hyphen}+{quote character}?
{open parenthesis}?{quote character}?
{capitalized or all-capital word, or number}
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Figure 2: A sample pattern for sentence-end detection, written in a generalized regular expression syntax

Word pronunciation

From studies of the behaviour of “general” letter-to-sound rules it is apparent that other strategies have to be employed to cater for name pronunciation [1], to handle derivatives, inflexions and words formed through compounding, and also to handle frequently occurring clusters of words (collocations). In working on our new TTS development system, we have chosen an approach where emphasis is moving towards lexical look-up. The main reason for this is that it is very difficult (and therefore unsuitable) to capture all the “lexicalities” that are apparent in the language using general letter-to-sound rules. This is in agreement with for instance Coker et al. [2], who argue that dictionary-based approaches are inherently capable of achieving higher accuracy than letter-to-sound rules. A lexicon can provide much more information than just the word pronunciation, e.g. syntactic and semantic information, and is also capable of generating multiple hypotheses, as is the case when dealing with homographs. A lexicon can also provide alternative transcriptions for the same word, for situations where either extremely clear pronunciation or a more reduced pronunciation is desirable. The first option is useful for emphasizing words or phrases (e.g. when paraphrasing), or reading modes where high intelligibility is required, the second may be used for unemphasized words in a more casual reading style.

Of course one will still need fall-back strategies, including letter-to-sound rules, but other approaches suggested in the literature should also be considered, e.g. pronunciation by analogy [2, 3].

Lexicon The Swedish lexicon was developed at Chalmers Institute of Technology, Department of Information Theory [7]. It contains, in its latest edition, approximately 116,000 entries (including abbreviations, acronyms, common compounds and proper names). Additionally, a lexicon of 30,000 Swedish proper names [1] is available to the system. The lexicon is lemmatized, i.e., each head-word is associated with an inflexion rule from a set of approximately 500 rules that is capable of generating all possible inflexions of the word. Each word is listed in the lexicon with its part-of-speech specification, inflexion code, phonetic transcription(s), information regarding its morphological make-up, as well as derivational and compound information, as exemplified in Table 1 by the word “maskerad” (*masked, masquerade*).

A lemmatized lexicon offers significant advantages over a lexicon based on surface forms and morphological truncation, since it maintains the relationship between the

orthography	maskerad	maskerad	maskerad
part-of-speech	verb	noun	adjective
paradigm-info	tr, prf.p, nn, sing ind, nom	nn, indef, sing, nom	po, nn, indef, sing nom
syllable	ma-ske-rad	ma-ske-rad	ma-ske-rad
transcription	/ma'ske:rad/	/maske'ra:d/	/ma'ske:rad/
derivation	-	-	maskera+ad
compound	-	-	-

Table 1: The information supplied by the lexicon when analyzing the word *maskerad* (*masked*, *masquerade*)
tr: transitive, prf.p: perfect participle, nn: non-neuter, sin: singular, ind: indirect, nom: nominative, po: positive

inflected forms and the head-word, a prerequisite in semantic/pragmatic processing [8]. The lexicon can also handle compound words, either by explicit listing or by algorithmic generation. This is a necessity in languages like Swedish and German where compounds are written without indication of internal word boundaries. The lexicon is searched using an efficient tree-search algorithm, with demand driven expansion into inflected forms.

The performance of the lexicon was investigated using a corpus of 100,000 words (10,000 unique surface forms) extracted from some of the Swedish texts of the SUC [4]. The coverage varied between 96 and 99% on these texts. About 20% of the look-ups generated multiple hypotheses. Cases of failed look-ups included unknown proper names, multiple compounds, special terminology, non-standard oldfashioned spelling, misspellings, foreign words etc.

Upper/lower case significance When the distinction between upper and lower case is significant, it is often possible to utilize that information in order to reduce the number of hypotheses delivered by the lexicon. Capitalized words, for example, generally relate to proper names, while upper case words often refer to acronyms (e.g. IBM, GATT) or possibly emphasized words. A sentence of all upper case or capitalized words may be a title or a headline. Consider, for example, the Swedish words "hans" and "Hans" (*his*, *Hans* or *Hans's*), which have different phonetic realizations. As long as "hans" or "Hans" do not appear as the first word in the sentence, the ambiguity can be resolved by utilizing case information.

Given that in most orthographic systems a word can be either "Capitalized" (c), "UPPER CASE" (u) or "lower case" (l), case is significant if, within a contextual window, words of more than one of the types c, u or l exist. The window could be set either to a fixed number of words or to the sentence established by a sentence-end detection algorithm.

Collocations The identification of collocations, i.e. recurrent combinations of words as they appear in context, is something which could considerably increase the naturalness of synthetic speech. In human speech, collocations act as prosodic units and are subject to a higher degree of reduction and internal coarticulation than they would be, had they been ordinary, separate words. In

i fall *conjunction* /i'fall/ (in case)
i går *adverb* /i'gå:r/ (yesterday)
av och till *word group* /'a:vå'till/ (occasionally)
Motåla ström *place name* /mo:tåla'strömm/

Figure 3: Examples of collocations

accordance with Hedelin and Huber [7], we have chosen a lexical approach for handling collocations. Currently, there are approximately 200 common collocations in the lexicon, not counting particle verbs or reflexive verbs, which amount to approx. 2,500. Some of these are shown in Fig. 3.

It is very common that TTS systems treat "space" as a word delimiter. This is obviously essential to avoid when dealing with collocations (and also in some other cases, which are mentioned below).

In some cases, syntactic knowledge is needed to differentiate between an interpretation of a group of words as a collocation or as the words as such:

Buren som papegojan sitter i går inte att stänga.
The cage that the parrot is sitting in, can't be closed.
Jag var på bio i går.
I went to the cinema yesterday.

In the first sentence, "i" is the preposition *in* and "går" means *can*, and there is a major syntactic (and prosodic) boundary between the two. It is only in the second case that "i går" should be treated as a collocation.

Abbreviations Abbreviations are also dealt with primarily using a lexical approach. This has been amended with a grammar of different typing conventions that are commonly used, so that only one generic form of the abbreviation is stored in the lexicon, e.g. "t_ex.", and this expands to all the possible ways of typing it: "t ex", "t. ex.", or "t.ex.". Note that for two of the cases we have the same kind of "multi-word" problem that appeared in connection with collocations.

Acronyms Letter strings in which all letters are in upper case (or *could* be, according to the case significance ks), are dealt with in a separate module.

The pronunciation of initialisms and other acronyms seems to a great degree to be lexicalized, and is treated accordingly in our system. However, since this is a very productive way of forming new "words", algorithms dealing with new acronyms are needed. Most alphabetic languages share the problem of determining whether a potential acronym should be syllabified and spoken as a "normal" word, or whether it should be spelled out. Determining this will require access to knowledge about graphotactics, phonotactics and syllabic structure. In Swedish, even if you choose the passive strategy of spelling out all acronyms not found in the lexicon, there will still be the problem of assigning the correct accent to the reading, either "accent 1" on the last syllable or compound accent.

Numeric expressions The interpretation of numeric expressions is highly context-dependent. The expression could be read as one number, digit-by-digit, or according

to some other convention. Human interpretation of digit strings is controlled primarily by pragmatic and semantic factors. In certain applications, e.g. dialogue systems, pragmatic information may well be available to the TTS system. In the case of unrestricted text-to-speech, inferences regarding the pragmatics will have to be made from clues given in the text.

The textual clues can exist in the form of the presence of certain words, such as "p.m.", "phone", "zip-code" etc., or the existence of certain syntactic configurations. Each type of numeric expressions has its own "grammar", which can be used to further reduce the number of hypotheses supplied.

Syntactic analysis

Syntactic analysis is present in most TTS systems, ranging from "island parsing" in early systems to the sophisticated techniques available today, including both knowledge-based and statistical approaches. Within the scope of this paper, the syntactic parser serves primarily as a disambiguator, but it is important to remember that, in general, syntactic parsing is useful for determining sentence-level prosody and it is also indirectly a prerequisite for higher levels of analysis (pragmatic, semantic). We use a top-down parsing mechanism with a definite clause grammar.

Lexico-semantic analysis

Present day TTS systems do not generally incorporate semantic or pragmatic analysis. It is often considered too complex and too strongly related to the particular application domain to be feasible. However, since many future TTS applications are likely to be domain-specific, it is interesting to study ways of utilizing such information. The input will in such applications consist of text restricted to the particular application domain, and thus allow for some kind of semantic or pragmatic processing. We have investigated one important type of semantic processing which involves the tracking of coreferential relations [8]. This allows one to mark words and expressions as representing "New" or "Given" information, which is necessary in order to assign suitable prosody to an utterance. This module relies heavily on the availability of lexical information, both for relating inflected, compounded and derived forms of a word to its lexical form and for modelling semantic relationships such as synonymy, hyponymy and part/whole relations.

DISCUSSION

The purpose of this work has been to improve the way in which TTS systems cope with text analysis and word pronunciation. The suggested approach involves dividing existing knowledge into modules, and letting these knowledge sources share data with each other only via a well-defined data structure. This offers several advantages, one being that the modularity serves the purpose of reducing the number of degrees of freedom, making knowledge development and (not least importantly)

knowledge maintenance easier. Another main benefit of the proposed scheme is that ambiguities of a variety of kinds can get a satisfactory resolution. This is the indirect result of the combination of using an approach handling multiple hypotheses and letting the individual modules incrementally add to the solution. This way, decisions do not have to be made until sufficient evidence has emerged, on which to base the decisions. Another enhancement regarding the system design is that different modes (e.g. different reading modes) can be implemented by simply changing the knowledge content in single modules or by changing the control structure. In a similar way, adaptation to different domains can be made through exchanging entire modules.

More work and research is obviously needed in each of the different research fields touched upon in this article. In particular, tools for tuning the control structure and the knowledge contents of different modules to particular text domains are needed. Good measures for determining the best choice among competing hypotheses are also required.

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